EE 372B — Custom PCB Design Report

Edison Altamirano Francisco Hernandez-Gardea Sahan Samarakoon

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1 Schematic and Layout

1.1 Schematic

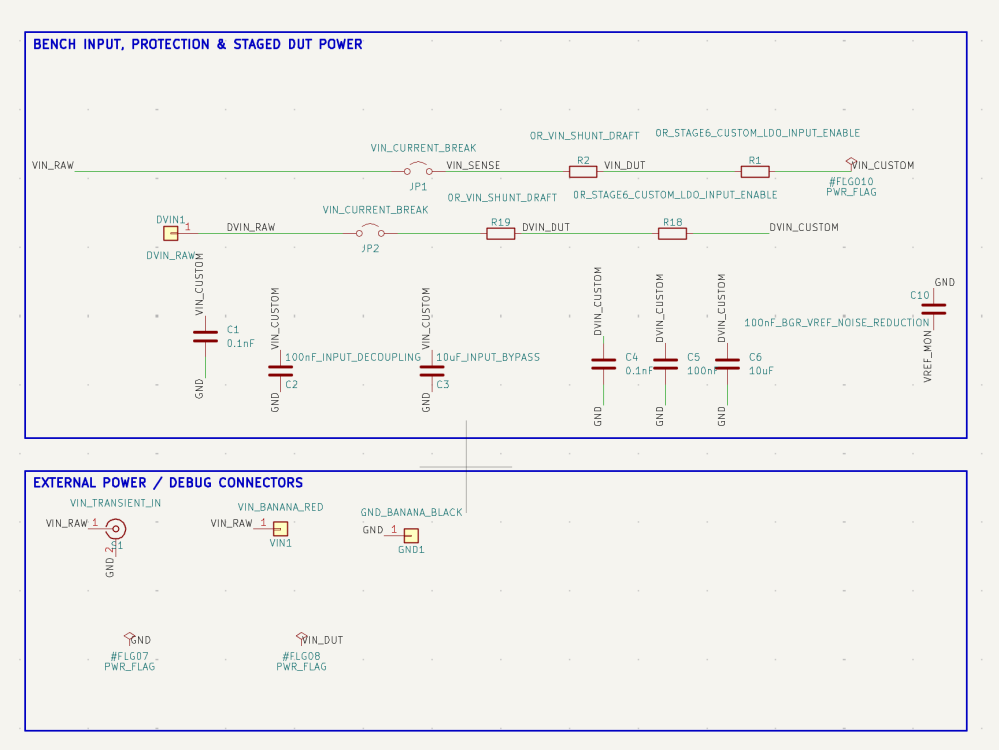


Figure 1: Schematic of Power Rails

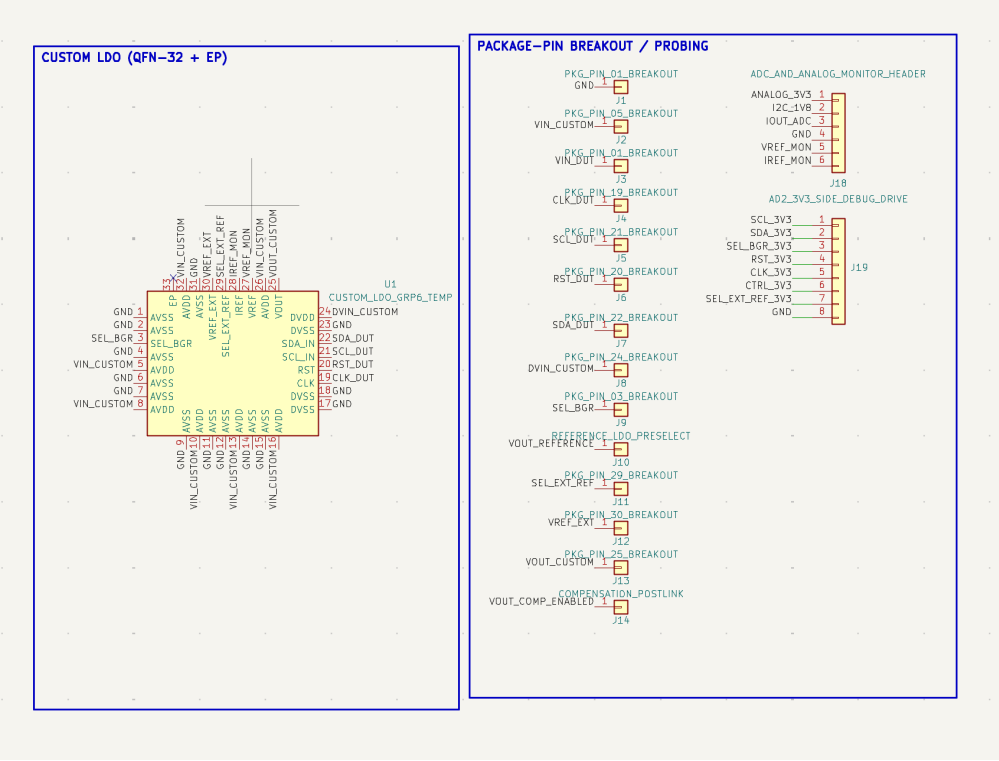


Figure 2: Schematic of IC and Test pins

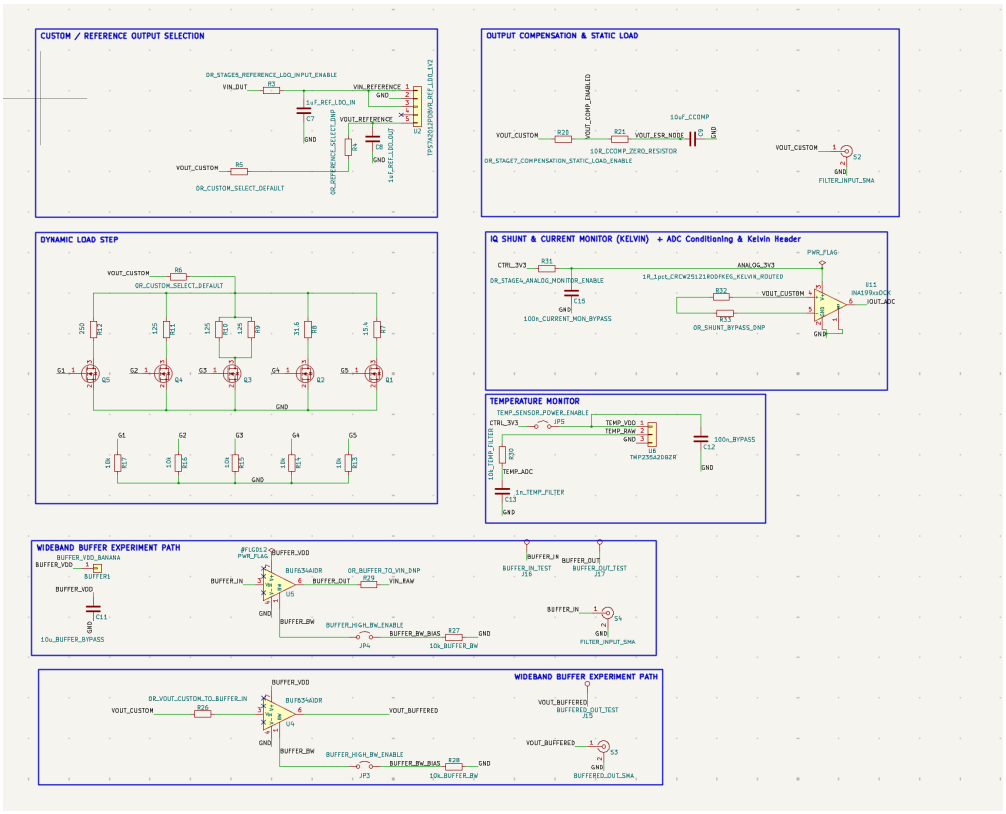


Figure 3: Schematic of Load side components

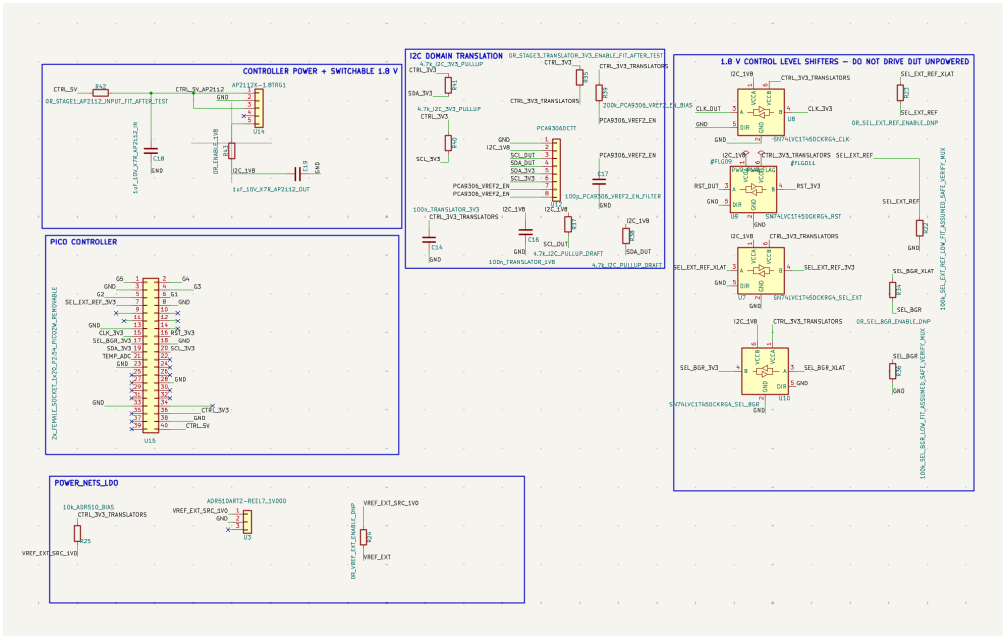


Figure 4: Schematic of I2C and MCU components

1.2 Layout

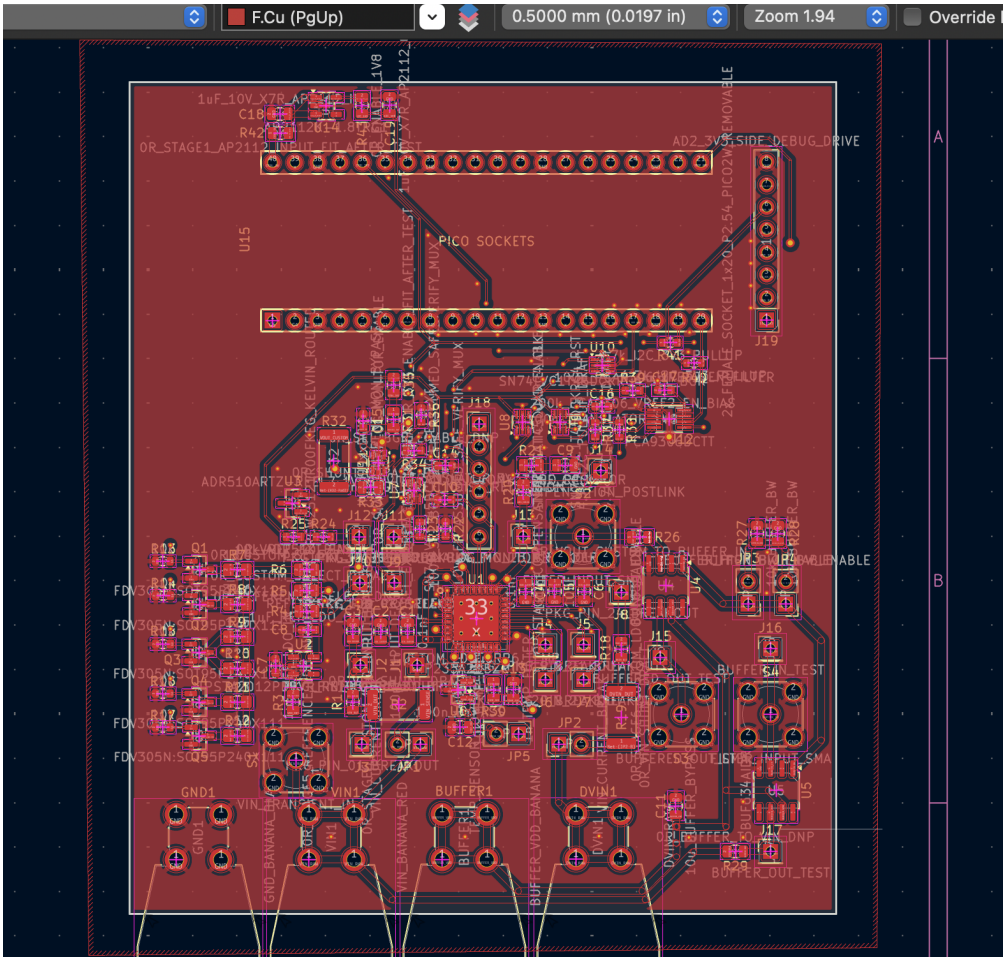


Figure 5: Top Layer Layout

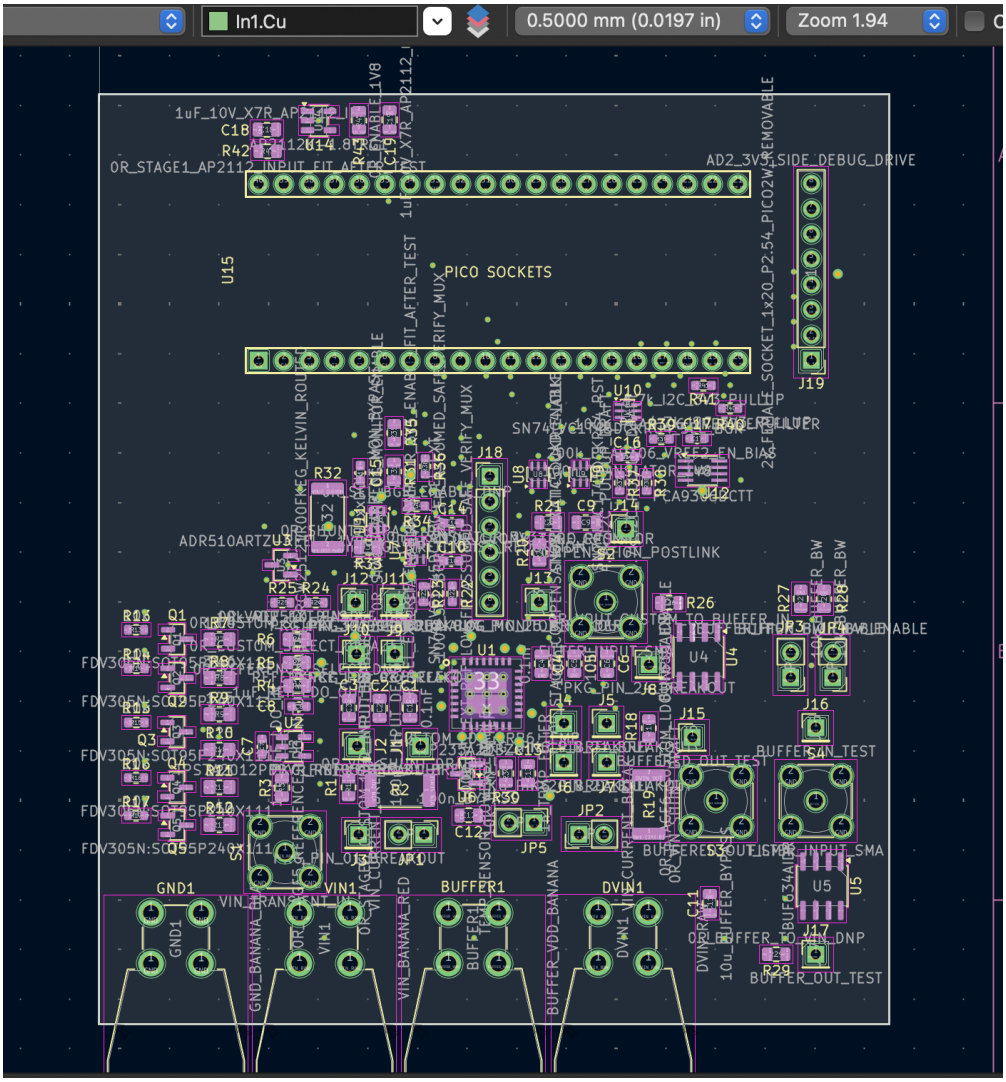


Figure 6: Second Layer Layout

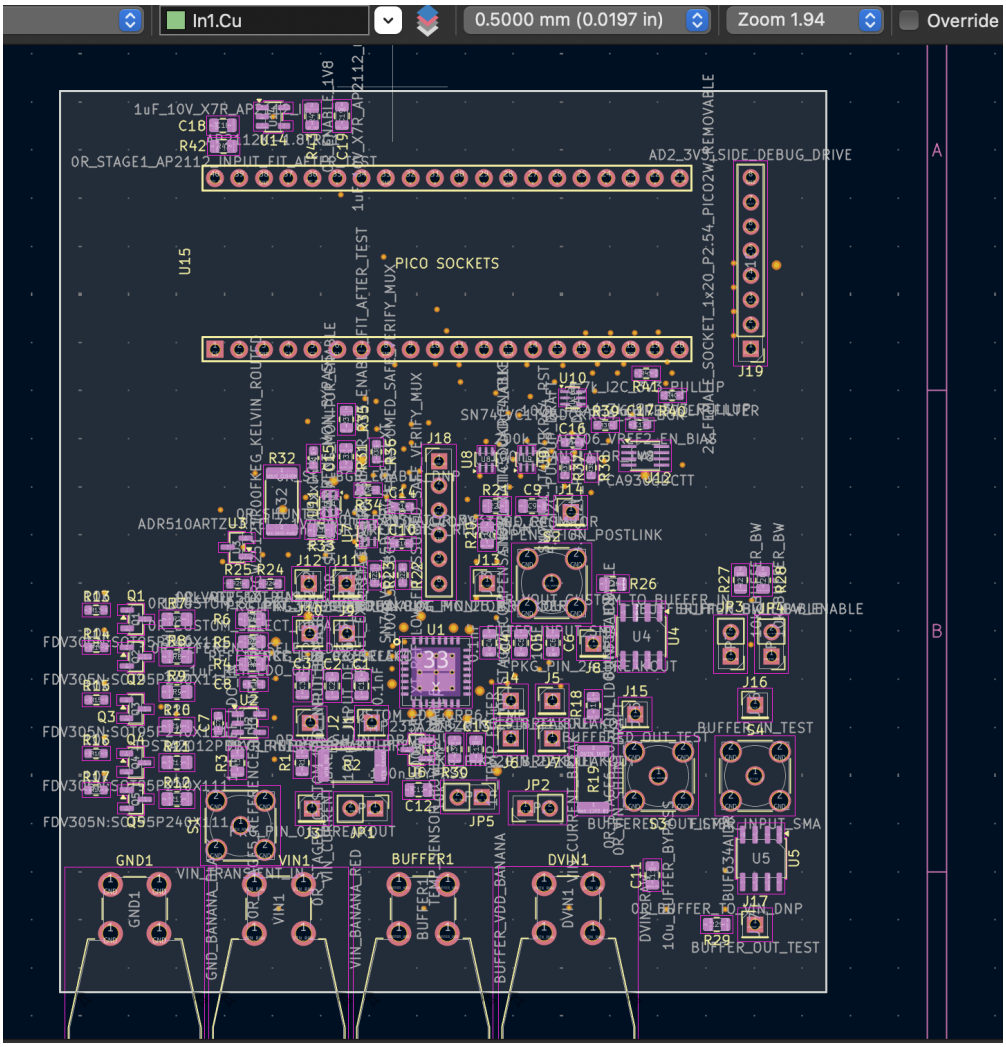


Figure 7: Third Layer Layout

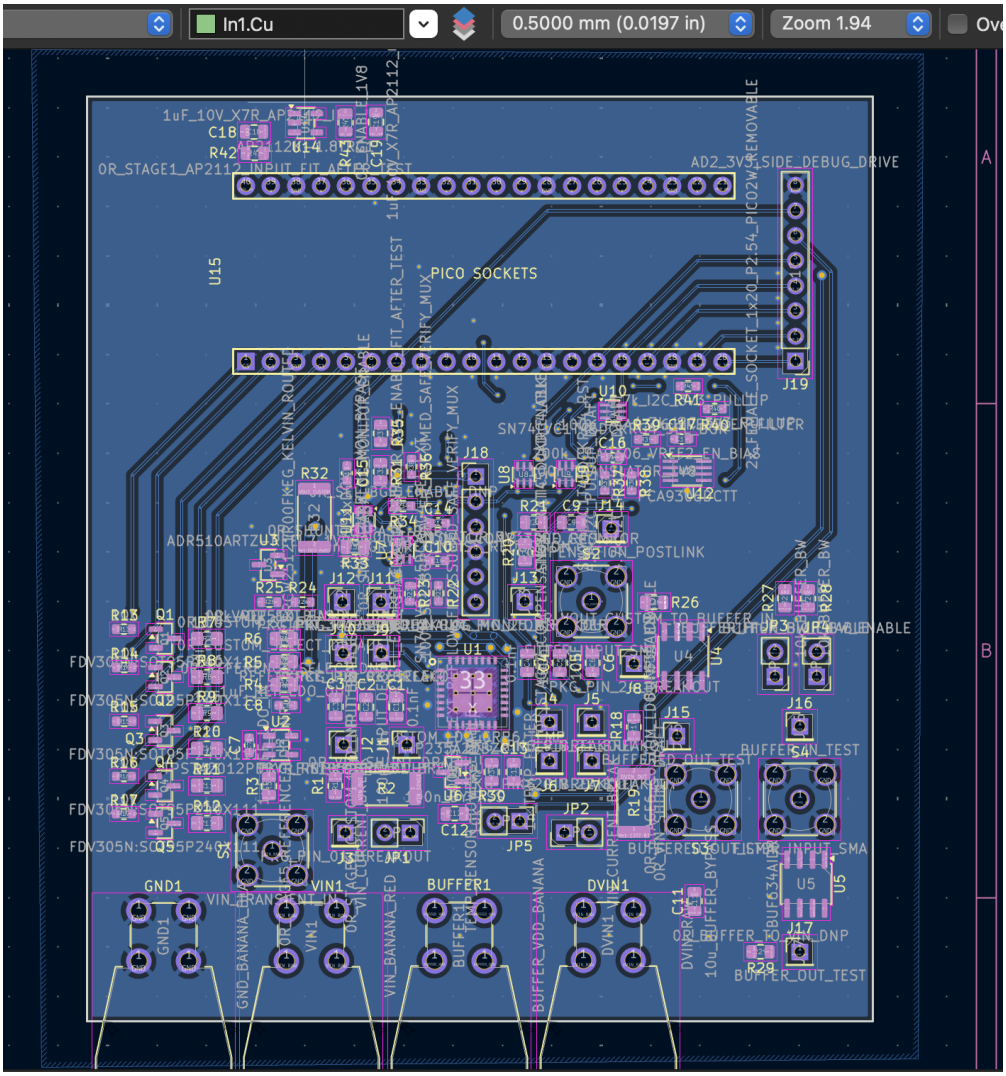


Figure 8: Bottom Layer Layout

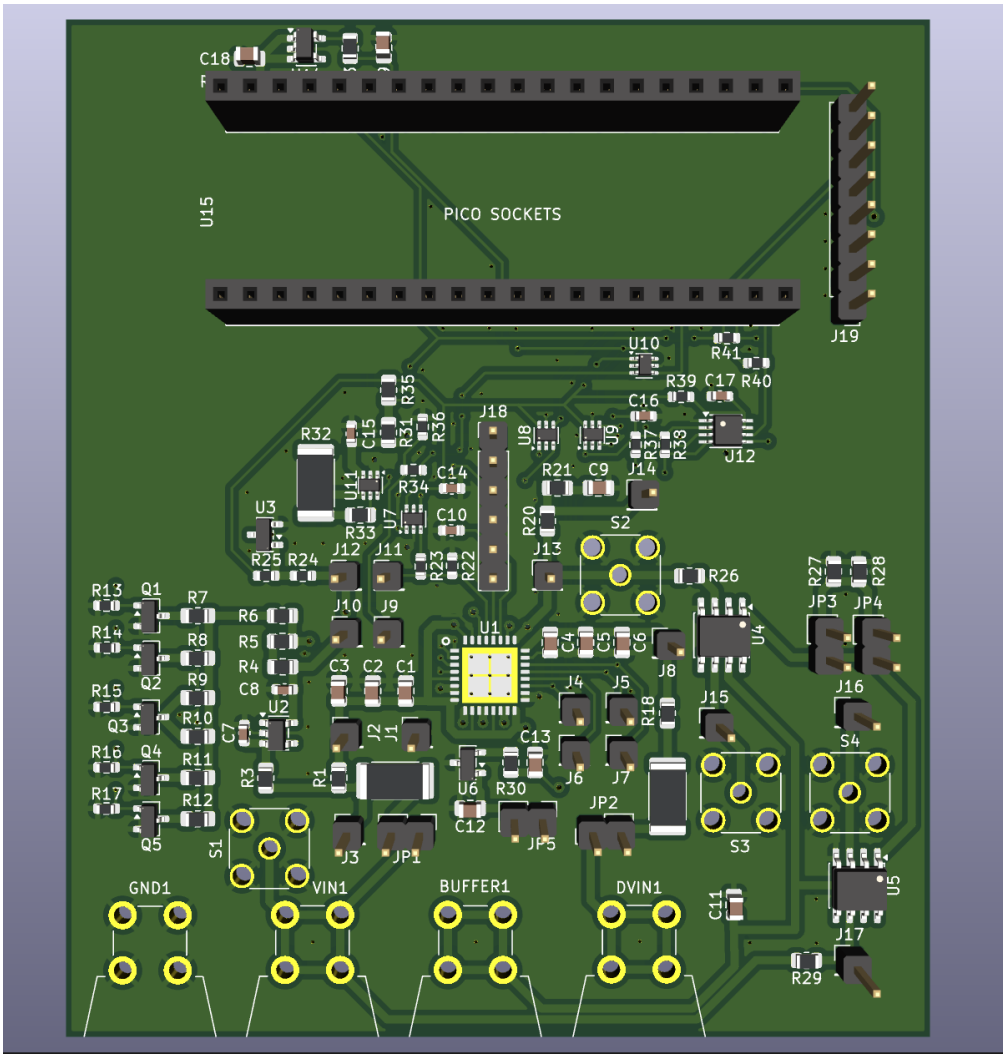


Figure 9: 3D View of PCB

2 Pin-Map and Footprint Verification

Table 1: QFN-32 + EP Footprint Dimensions

Feature	**Value**
Signal pad size	`0.35 mm × 0.8 mm`
Pad pitch	`0.65 mm`
Exposed-pad (EP) land	`5.1 mm ×5.1 mm`
EP thermal vias	`9 vias, 0.3 mm drill`
Courtyard	`7.55 mm × 7.55 mm`

Table 2: Package Pin-Map Verification

Pkg Lead	Pad #	Pad Type	Cadence Net	KiCad Pin Name	Board Net	Checked
1	1	PVSS2AC	VSS	AVSS	GND	SS
2	2	PVSS2AC	VSS	AVSS	GND	SS
3	3	PDB3AC	SEL_BGR	SEL_BGR	SEL_BGR	SS
4	4	PVSS2AC	VSS	AVSS	GND	SS
5	5	PVDD3AC	VDD	AVDD	VIN_CUSTOM	SS
6	6	PVSS2AC	VSS	AVSS	GND	SS
7	7	PVSS2AC	VSS	AVSS	GND	SS
8	8	PVDD3AC	VDD	AVDD	VIN_CUSTOM	SS
9	9	PVSS2AC	VSS	AVSS	GND	SS
10	10	PVDD3AC	VDD	AVDD	VIN_CUSTOM	SS
11	11	PVSS2AC	VSS	AVSS	GND	SS
12	12	PVSS2AC	VSS	AVSS	GND	SS
13	13	PVDD3AC	VDD	AVDD	VIN_CUSTOM	SS
14	14	PVSS2AC	VSS	AVSS	GND	SS
15	15	PVSS2AC	VSS	AVSS	GND	SS
16	16	PVDD3AC	VDD	AVDD	VIN_CUSTOM	SS
17	17	PVSS2AC	VSS	DVSS	GND	SS
18	18	PVSS2AC	VSS	DVSS	GND	SS
19	19	PDB3AC	CLK	CLK	CLK_DUT	SS
20	20	PDB3AC	RST	RST	RST_DUT	SS
21	21	PDB3AC	SCL_IN	SCL_IN	SCL_DUT	SS
22	22	PDB3AC	SDA_IN	SDA_IN	SDA_DUT	SS
23	23	PVSS2AC	VSS	DVSS	GND	SS
24	24	PVDD3AC	VDD	DVDD	DVIN_CUSTOM	SS
25	25	PDB3AC	VOUT	VOUT	VOUT_CUSTOM	SS
26	26	PVDD3AC	VDD	AVDD	VIN_CUSTOM	SS
27	27	PDB3AC	VREF	VREF	VREF_MON	SS
28	28	PDB3AC	IREF	IREF	IREF_MON	SS
29	29	PDB3AC	SEL_EXT_REF	SEL_EXT_REF	SEL_EXT_REF	SS
30	30	PDB3AC	V_REF_EXT	V_REF_EXT	V_REF_EXT	SS
31	31	PVSS2AC	VSS	AVSS	GND	SS
32	32	PVDD3AC	VDD	AVDD	VIN_CUSTOM	SS
EP	33	EP	N/A	EP	N/A	SS

3 Test Features

3.1 Rail Inventory

Table 3 summarizes the supply and ground domains brought out from the custom IC. The analog and digital supply domains are kept separate on the PCB to allow independent characterization.

Table 3: Custom IC Rail Inventory

Board Rail	IC Domain / Pins	Nominal	Access / Isolation	Local Decoupling
VIN_CUSTOM	Analog supply (AVDD), pins 5, 8, 10, 13, 16, 26, 32	1.8 V	Independent analog supply path with current-break jumper and 0 Ω series link. Rail may be swept below 1.8 V for line-regulation and dropout measurements.	100 pF, 100 nF, and 10 μF
DVIN_CUSTOM	Digital supply (DVDD), pin 24	1.8 V	Independent digital supply path with current-break jumper and 0 Ω series link; may remain at 1.8 V while VIN_CUSTOM is swept.	100 pF, 100 nF, and 10 μF
GND	Analog and digital grounds (AVSS: 1, 2, 4, 6, 7, 9, 11, 12, 14, 15, 31; DVSS: 17, 18, 23)	0 V	Common solid ground plane and ground test points	—

The exposed pad, pin 33, is maintained as the separate EP_NET connection and is not treated as an independent supply rail.

3.2 Specification Measurement Access

Table 4 summarizes the PCB features used to characterize the custom LDO and on-chip reference circuits.

Table 4: Specification-to-Board Measurement Feature Mapping

Specification	**Relevant Net(s)**	**Board Measurement Feature**
DC output voltage	VOUT_CUSTOM	Direct custom-LDO output test point (TP37); additional monitored output available through the VOUT measurement path.
Line regulation / dropout	VIN_CUSTOM, VOUT_CUSTOM	Sweep the analog bench supply while monitoring VIN_CUSTOM at TP28 and the custom LDO output. DVIN_CUSTOM can remain independently powered at 1.8 V.
Load regulation	VOUT, VOUT_EXTERNAL	J110 external-load terminal block; 1 Ω Kelvin current shunt and INA180 current monitor provide load-current measurement.
Load transient response	DYN_GATE_DRIVE, VOUT	J102 LOAD_STEP_DRIVE SMA drives the MOSFET load-step circuit; monitor VOUT using TP45 or J114.
PSRR	VIN_RAW/VIN_CUSTOM, VOUT	J59 VIN_TRANSIENT_IN SMA provides AC input injection; output ripple is observed at J114 VOUT_SCOPE_NOISE.
Output noise	VOUT	J114 VOUT_SCOPE_NOISE SMA provides a coaxial connection for low-noise output measurement.
Quiescent current (I_Q)	VIN_CUSTOM	Remove the analog VIN current-break jumper and insert a DMM or SMU in series with the DUT analog supply.
Reference voltage	VREF_MON	VREF probe access through J68 and the analog monitor header J111; package-pin breakout also provides direct access.
Reference current	IREF_MON	IREF probe access through J68 and the analog monitor header J111; package-pin breakout also provides direct access.
Temperature dependence	VOUT, VREF_MON, TEMP_ADC	On-board temperature monitor can be read by the Pico and correlated with VOUT and VREF measurements.
Digital control / trim	SDA_DUT, SCL_DUT, CLK_DUT, RST_DUT	Pico controller and 1.8 V level-translation circuitry exercise the digital interface; all DUT pins are additionally broken out for probing.

4 DRC and ERC Reports

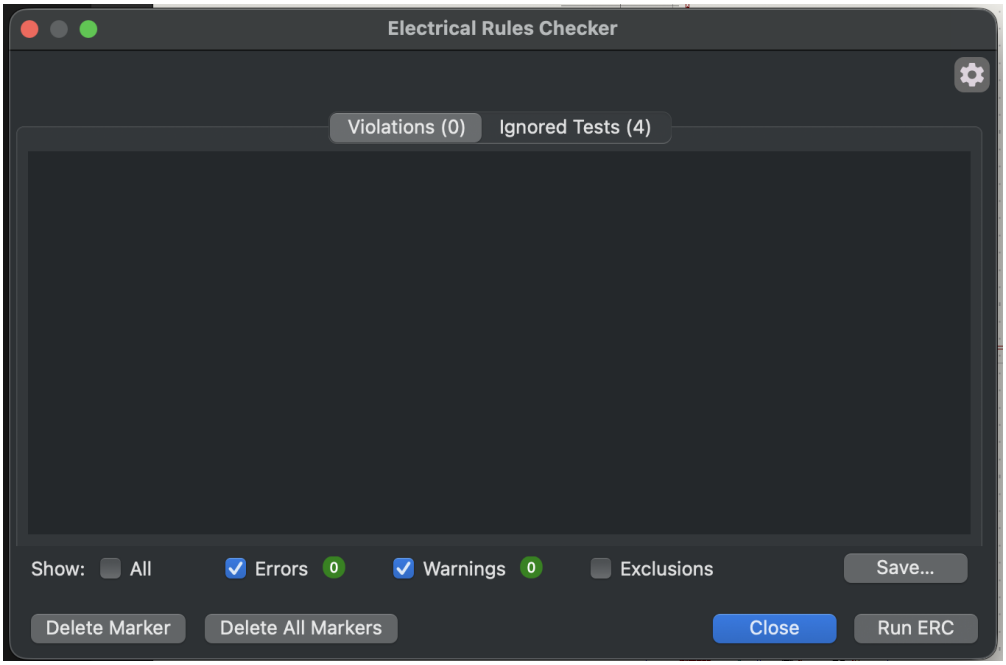


Figure 10: ERC Report with 0 errors

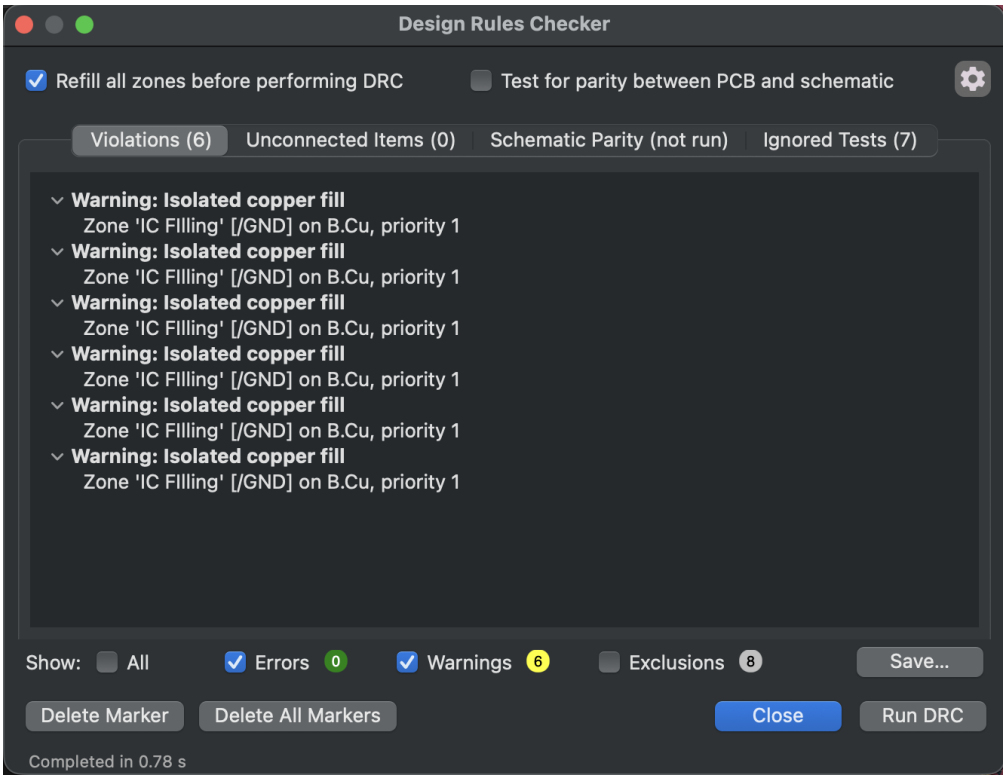


Figure 11: DRC Report with 0 errors